

# ECE 210: Circuit Analysis I

## Spring 2025 Course Syllabus

### Course Information

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| <b>Description:</b>   | DC and AC steady-state circuit analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Topics:</b> | Current, Voltage, Energy, and Power<br>Kirchoff's Current and Voltage Laws, Ohm's Law<br>Resistance Combination, Voltage and Current Division, Wye-Delta Transformation<br>Nodal Analysis, Mesh Analysis, Superposition, Source Transformation<br>Thevenin/Norton Equivalent Circuits, Maximum Power Transfer<br>Operational Amplifiers (Op Amps)<br>Capacitors and Inductors, Sinusoids, Phasor Analysis<br>Instantaneous and Average Power, RMS Values, Complex Power |
| <b>Prerequisites:</b> | A grade of C or better in MATH 150.                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Modality:</b>      | Classes are in-person and meet in the Sciences East building, room 2214 on Tuesday and Thursday from 9:30AM to 10:45AM.                                                                                                                                                                                                                                                                                                                                                 |
| <b>Textbook:</b>      | <b>Fundamentals of Electric Circuits (6th Edition)</b> , Alexander and Sadiku. CRC Press, ISBN 978-0078028229.                                                                                                                                                                                                                                                                                                                                                          |

### Instructor Information

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| <b>Name:</b>          | Julian Rene Cuellar Buritica, Ph.D. candidate ( <a href="mailto:jucuell@siue.edu">jucuell@siue.edu</a> )                                                                                                                                       |
| <b>Office:</b>        | EB 3145                                                                                                                                                                                                                                        |
| <b>Office Hours:</b>  | Tuesdays and Thursdays from 2:30PM - 4:30PM, or by appointment                                                                                                                                                                                 |
| <b>Communication:</b> | We will discuss any ongoing schedule changes and plans for the course in class; it is therefore imperative that you attend! I will use your SIUE email address for the primary means of communication outside of class; please check it often. |

### Supplemental Instruction (SI)

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| <b>Leader:</b> | Shaun Pacatte, ECE student ( <a href="mailto:spacatt@siue.edu">spacatt@siue.edu</a> ) |
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This course participates in Supplemental Instruction (SI) at SIUE, which is a peer-led group study program for historically difficult courses. SI Leaders—students who took this course on campus and received a good grade—offer two to three group study sessions outside class per week to help students discuss important concepts, develop study and test taking strategies, and focus on how to learn with what to learn.

### Class Policies

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| <b>Attendance:</b> | You are expected to be present and seated before the start of class and remain present until the class period is over. If you miss a class, you are required to let me know, preferably before the start of class, but absolutely by 6PM of the same day. Failure to notify me of your absence will double the penalty to your <i>Participation &amp; Professionalism</i> points. In the event of an emergency, exceptions to this notification policy will be handled on a case-by-case basis. Also, as per SIUE's Academic Policies and Requirements, "... failure to actively participate [in a class] may result in |
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a reduction or removal of financial aid."

Attendance will be taken using StarFish for each class. If you are late for class, I will mark you absent. I am using StarFish so that if you are persistently absent, somebody from the Office of Retention and Student Success may follow up with you and can help work through any issues you may be experiencing.

More than three unexcused absences throughout the term puts you at risk of being dropped from the class as per SIUE's [Course Enrollment Policies](#). However, this is not an acceptable method of dropping this class should you decide to do so.

You will not be penalized due to absences related to career advancement (e.g. advisement, job fair, interview, etc.), provided you notify me before the missed class period.

Students who must miss class for official university business must follow SIUE's [Student Academic Standards and Performance](#) policy:

*Students who are absent from class due to official University Business (e.g., athletic events, music, theater, dance productions, academic field trips, Student Senate) will be excused for up to 15 percent of the scheduled contact hours of their classes. Such an excuse is contingent, however, on the student and the sponsoring unit informing the instructor(s) and advisor in writing at least one week in advance of the anticipated absence.*

**Reading:**

Reading assignments may be made occasionally throughout the semester. It is essential that you read the assigned sections by the specified date. Some homework problems require you to know and apply concepts that are covered in the reading assignment but might not be discussed thoroughly in class due to large amount of material to be covered.

**Assignments:**

Assignments and class-related information (e.g. homework, quiz dates, SI sessions, etc.) may be posted to the class' Blackboard site or communicated via email. It is your responsibility to check to stay up-to-date on any new information that might impact your grade.

**Homework:**

A list of homework problems will be assigned for each topic and subsequent quiz. Although you will not be turning them in, you should ***work all of the problems*** in the homework set because every question on the associated quiz will be a modified version of one of those homework problems. In addition, you are allowed to look at your homework while taking the quiz, subject to the conditions outlined in the *Quizzes* policy section.

**Exams:**

The only exam in this class is an optional comprehensive final exam given during finals week.

**Quizzes:**

Timed quizzes are given frequently, with each quiz covering a specific topic and homework set. The total number of quizzes given in a particular semester will vary, but nominally, we expect one per week for a total of 15. However, I reserve the right to add or remove scheduled quizzes. I also reserve the right to assign take-home quizzes that will be done as homework and turned in for a quiz grade. In addition, inclement weather could impact the quiz schedule.

While taking a quiz, you are not allowed to look at your class notes, textbook, or online resources; however, you can look at your homework solutions during the quiz, provided you adhere to the following conditions:

- The material only contains homework problems from the assigned homework set
- It is your own handwritten work and not a printout or copy of someone else's work
- It is either written in a spiral-bound notebook or stapled together *before* the start of class
- You *must* hold it out for me to look at before I will accept your quiz

In general, make-up quizzes are not given. If you are unable to take a quiz for an excusable reason (e.g. an urgent medical situation), then that quiz will not be included in your total quiz

grade. However, this effectively makes your other quizzes weigh more heavily in your total quiz grade. If you do not have an excusable reason, then the missed quiz will be recorded with a score of zero.

**Mathematical Tools:** Many of the calculations needed in this class require the solving of algebraic and trigonometric equations, sometimes with three (or more) variables. Later in the semester, these variables will be complex-valued. Although calculus is not used extensively, you may be required to use it on occasion. Since you have passed the prerequisite math classes, what is important to me is that you can formulate the equations that need to be solved rather than getting mired in the details of solving them. This means that **you must have a suitable math solver for the homework, quizzes, and final exam**. I will potentially be using both *WolframAlpha* and *MATLAB* for calculations during class, and I encourage you to use both of these too; however, you may also use any suitable calculator (e.g. TI-Nspire), but it is your responsibility to figure out how to use it.

**Laptops, Tablets, & Phones:** You are allowed to use a laptop, tablet, or phone during class, but only if they are used in a manner that is not disruptive to those around you.

I may use both *WolframAlpha* and *MATLAB* for my calculations in class. If you have a laptop, I encourage you to bring it to class and use it to replicate my calculations so you can become proficient with these tools.

You can also use your laptop while taking a quiz or exam, subject to the following conditions. Note that violating any of these will result in you receiving a zero on the quiz or exam:

- You only use it for calculations with either *WolframAlpha* and/or *MATLAB*
- You do not look at any online references, including Blackboard, etc.
- You do not use it to communicate with anyone
- You make an effort to keep your screen from being seen by people sitting next to you
- You do not try to hide your screen from me as I walk around the class
- You immediately show me all applications that you have running if I ask to see them and make no attempt to close or hide them while doing so

If you do not have a laptop or if you forget to bring it for a quiz, I might permit you to use a tablet or phone provided you get my approval for the specific apps or websites that you will be using.

**IA Class support:** Our university is providing you access to CircleIn for free to help you study and collaborate with each other and CircleIn also has an AI Tutor that helps you get support and feedback at anytime. Go to [CircleInApp.com](https://circleinapp.com) to login, and you search for CircleIn in the app stores to download the app. We all want you to do well in my course, and CircleIn is a free resource to help you. Please give it a try early and often.

**ACCESS:** Students needing accommodations because of medical diagnosis or major life impairment will need to register with Accessible Campus Community and Equitable Student Support (ACCESS) and complete an intake process before accommodations will be given. Students who believe they have a diagnosis but do not have documentation should contact ACCESS for assistance and/or appropriate referral. The ACCESS office is located in the Student Success Center, Room 1270. You can also reach the office by e-mail at [myaccess@siue.edu](mailto:myaccess@siue.edu) or by calling 618-650-3726. For more information on policies, procedures, or necessary forms, please visit the ACCESS website at [www.siue.edu/access](http://www.siue.edu/access).

## Grading Policies

**Summary:** All grading will consider (but will not necessarily be limited to) use of correct theory or equation(s), proper application of theory or equation(s), mathematical correctness, reasonable

degree of accuracy, adequate labeling of sketches, organization, neatness, necessary assumptions, and correct conclusions. Basically, the points you earn for your solution are a measure of *how correct the entire solution is in both the approach used and the results shown*. Grading is not a simple checklist of points to be gained or lost; details are covered in the Partial Credit section.

**Quizzes:**

All quizzes are worth 10 points each. As stated earlier, we expect approximately 15 quizzes throughout the semester. Your two lowest quiz scores are excluded from your grade calculation *except for the last two quizzes of the semester*; the last two quizzes will count towards your total quiz score and *cannot be skipped*. This means that, if there are  $N+2$  quizzes given during the semester, your total quiz score is the sum of your best  $N-2$  quiz scores on the first  $N$  quizzes, plus your scores on the last 2 quizzes of the semester. To help illustrate this, suppose that 15 quizzes are given this semester. Your total quiz score would be equal to the sum of your 11 best scores on quizzes 1 through 13 plus your scores on quizzes 14 and 15.

**Final Exam:  
(Optional)**

Before finals week, I will compute your grade based on the *Non-Comprehensive Grading Scheme* and ask if you would like to keep it as your class grade and opt-out of taking the final exam. If you choose to do this, you must confirm this with me at least two days before the final exam begins. If you choose to take the final exam or do not opt-out in time and must take it, your class grade will be based on the *Comprehensive Grading Scheme* (shown below right). Note: if you take the final exam, you **will not** have the option of retroactively opting-out even if your score on the final causes you to have a lower grade than the *Non-Comprehensive Grading Scheme*.

**Participation &  
Professionalism:**

Being an active participant in the class and conducting yourself in a professional manner comprises 5% of your total class grade (i.e. half a letter grade). If you follow all instructions and requirements and do not have any unexcused absences, you will earn 2.5% of these points. To earn the remaining points, you must show active engagement in the class and being professionally responsible *throughout the semester*. The most effective ways are:

- occasionally asking or answering questions during lecture
- asking or answering questions in an SI session
- being present in an SI session

Points can also be lost for unprofessional behavior. The most common reasons are:

- unexcused absences (e.g. not medical, career, or university business)
- not notifying me of your absence as per the class attendance policy
- not following class policy or directions that are clearly written or are articulated in class
- asking a question without looking to see if the answer is readily available
- a single act of dishonesty will cause you to irredeemably lose all 5% for the semester

**Grading Questions:**

After work is returned, you have up to 48-hours to ask me questions about how it was graded or if you think there is a grading mistake. I really do want to hear about it; I just need it to be timely. But to be clear, there are no negotiations on “points,” but if there is a grading mistake, I really do want to know about it.

**Partial Credit:**

If your proposed solution to a particular problem clearly shows that you understand all the concepts involved in solving the problem, but you have made a minor, non-conceptual, error, such as misreading a dimension or failing to convert units correctly, then you will receive partial credit for your work subject to the rubric below. However, partial credit is not awarded for a proposed solution which does not demonstrate knowledge of all the concepts required to solve the problem.

*Metaphorically speaking, partial credit is awarded if the cake you bake is edible but not delicious; it is not awarded merely for putting the batter in the pan, but forgetting to turn on the oven, or for setting the oven at the right temperature, but then baking your shoes.*

— Mark Rossow, Ph.D

If, in answering any question, you state something that is incorrect, you will be penalized even if you have also put down the correct answer. To extend Dr. Rossow's metaphor, if you bake a beautiful and delicious cake, and your shoes at the same time, it will not be considered correct.

| Score Range | Description                                                                                        | Example                                                                                   |
|-------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| 10          | Everything is correct, including all thinking, solutions, and calculations                         |                                                                                           |
| 8-9         | All circuit analysis principles are applied correctly, but there are one or more minor math errors | Incorrect sign from mathematical operation                                                |
| 6-8         | There are minor circuit analysis errors                                                            | Incorrect sign from KVL or KCL                                                            |
| 4-7         | There are major circuit analysis errors                                                            | Adding V to I, etc.                                                                       |
| 0-4         | There is little or no demonstration of understanding                                               | The problem was not attempted or is mostly blank with only minimal, incorrect information |

## Grading Scheme

| Comprehensive Grading Scheme                                                                                    |            |        | Non-Comprehensive Grading Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |        |
|-----------------------------------------------------------------------------------------------------------------|------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| Type                                                                                                            | Points     | Weight | Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Points   | Weight |
| Quizzes: <i>Sum</i> (top N-2 scores on Q <sub>1</sub> to Q <sub>N</sub> ) + Q <sub>N+1</sub> + Q <sub>N+2</sub> | 10 each    | 70%    | Quizzes: <i>Sum</i> (top N-2 scores on Q <sub>1</sub> to Q <sub>N</sub> ) + Q <sub>N+1</sub> + Q <sub>N+2</sub>                                                                                                                                                                                                                                                                                                                                                                         | 10 each  | 95%    |
| Final Exam                                                                                                      | TBD        | 25%    | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | N/A      |        |
| Participation and Professionalism                                                                               | 10 total   | 5%     | Participation and Professionalism                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10 total | 5%     |
| Grading Breakpoints*:                                                                                           |            |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |        |
| A                                                                                                               | 100% - 90% |        | *The final grading scale may be adjusted based on the performance of the entire class. However, the scale listed above is a guarantee, that is, if you achieve over 90% for the course, you are guaranteed an A for the course. According to the Catalog, the following grading symbols are used by SIUE: A-Excellent, B-Good, C-Satisfactory, D-Poor, and F-Failure (see Academic Policies and Requirements).<br><b>The letter grades will strictly comply with these definitions.</b> |          |        |
| B                                                                                                               | 89% - 80%  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |        |
| C                                                                                                               | 79% - 70%  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |        |
| D                                                                                                               | 69% - 60%  |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |        |
| F                                                                                                               | 0% - 59%   |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |          |        |

## Joint Work, Outside Help, and Academic Misconduct

The ability to share common interests and ideas is a valuable tool for learning. I *encourage each of you* to discuss the homework problems with your fellow students and to seek whatever help you can find when you do not understand something. Obviously, **this does not apply to the quizzes or exams.**

I assume that all of my students are fundamentally honest and have a high level of integrity. However, if I have reason to suspect that you have had a non-trivial amount of assistance on any graded course work, I will challenge you to solve a

similar problem and ask you to explain your solution method as you work it. If you cannot solve the challenge problem or if you refuse to show and explain your methodology, you will receive no points for that work in question. Furthermore, if I find evidence that you have committed [Academic Misconduct](#) (some examples include: looking at information during a quiz or exam that is not permitted; communicating with anyone during a quiz or exam; consulting an expert on *Chegg Study*; you and another student consistently making the same uniquely-wrong mistakes), I will present my evidence to the Provost's Office and recommend sanctions.

## **University Policies**

### **Recordings of Class Content**

Faculty recordings of lectures and/or other course materials are meant to facilitate student learning and to help facilitate a student catching up who has missed class due to illness or quarantine. As such, students are reminded that the recording, as well as replicating or sharing of any course content and/or course materials without the express permission of the instructor of record, is not permitted, and may be considered a violation of the University's Student Conduct Code (3C1), linked here: <https://www.siue.edu/policies/3c1.shtml>.